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Interface Circuits for a Six-Phase Variable Reluctance Energy Harvester

A comparative study of rectification and power conditioning
topologies

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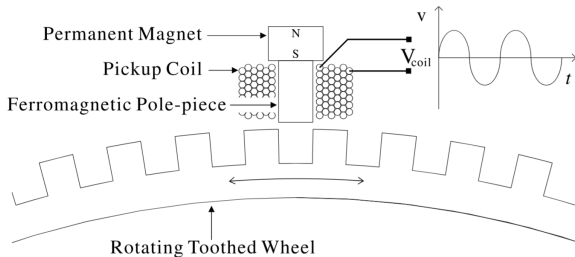
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Why energy harvesting?



- WSN and IoT nodes are everywhere, **collecting data** for many different purposes.
- Batteries are costly to maintain, have a finite life and are often **impossible to replace**, and they are bad for the environment.
- With **energy harvesting** the node powers itself from the energy already available around it.



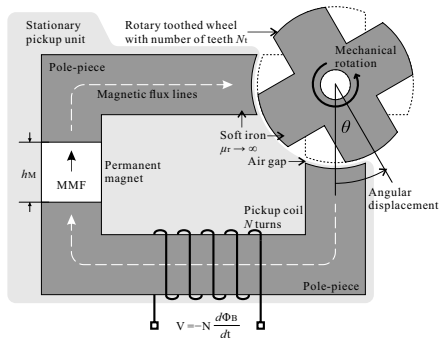
The variable reluctance principle



- Magnet and coil **stand still**; only the toothed wheel rotates.
- The passing teeth modulate the flux and induce an EMF.

Advantages

- No moving electrical parts
- Scalable to any shaft size
- Very small counter torque



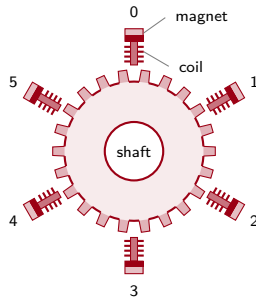
Pickup unit facing the rotating wheel.

The six-phase VREH



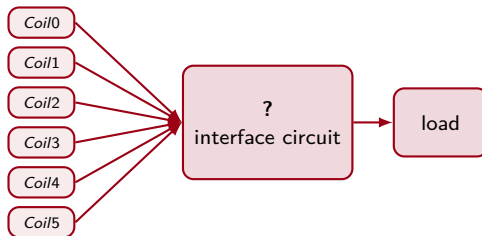
Six pickup units around the shaft, electrically 60° apart, evolved from railway prototypes into a design for large vehicles.

Target range **100 - 400 rpm** (20 - 80 km/h), with under 2V peak per phase, so the rectifier choice becomes critical.



Each unit is a permanent magnet over its pickup coil.

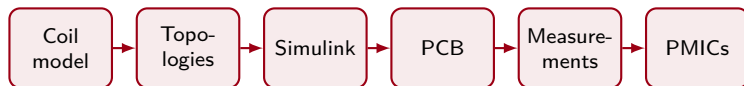
six coils, 60° apart



- Six phases give more power, but also **six time-shifted AC signals** that all have to be rectified.
- Earlier work only compared rectifiers for a **single phase**.

Research question

Which interface circuit extracts the most power, over the whole speed range?

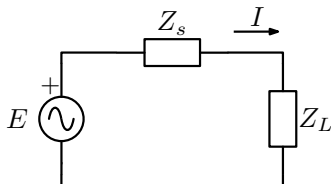
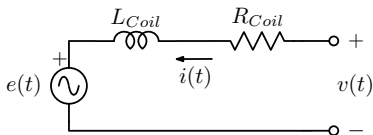


From theory to a real prototype, with **every topology measured on the same board** and on the same test bench.

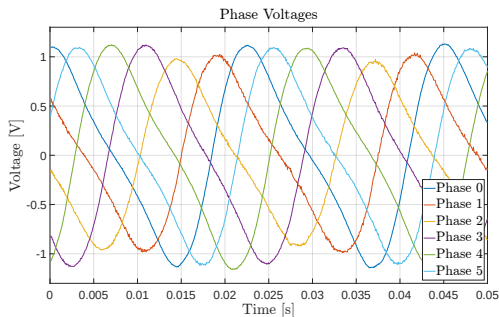
- Each coil is a source in series with R_{Coil} and L_{Coil} .
- At low speed a **resistive** load is enough.
- At higher speed $\omega_e L_{\text{Coil}}$ breaks the matching.
- A series capacitor C_m cancels it

$$C_m = \frac{1}{\omega_e^2 L_{\text{Coil}}}$$

valid at one frequency at a time.



The six real phase voltages



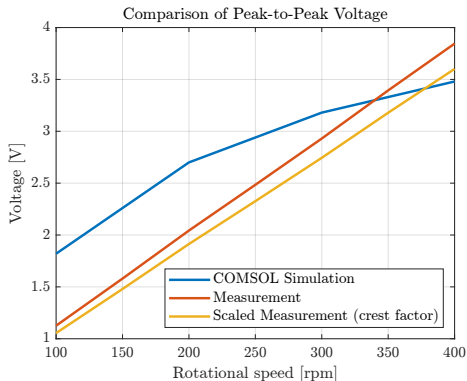
- Measured at 100 rpm.
- The 60° shift is clearly visible.
- **Not** clean sine waves, and the six units are not identical.

- RMS voltage computed against speed.
- Each source replaced by an **equivalent sinusoid**, scaled by the crest factor.
- Same power as the real waveform.

Every phase is then one source, delayed by 60° from the previous one,

$$e_k(t) = E_{pk} \sin\left(\omega_e t - k \frac{\pi}{3}\right)$$

with $k = 0, 1, \dots, 5$.



FEM, raw measured peak, crest-factor corrected.

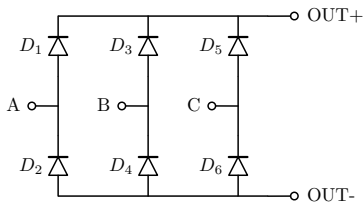
Strategy 1 - FWR bridges



Two three-phase groups ($k = 0, 2, 4$ and $k = 1, 3, 5$), each on a standard bridge.

- The **Star (Y)** gives a line voltage $\sqrt{3}$ times the phase EMF, so the two-diode drop costs less.
- The **Delta (Δ)** gives the coil EMF itself, with no gain but no floating node.

The Star should give the **higher open-circuit voltage**.



Two diodes always in the current path.

Strategy 2 - NVC bridges

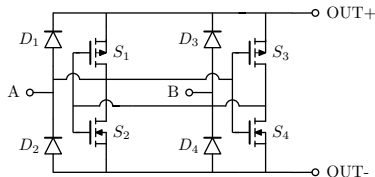


Three two-phase groups, pairing phases 180° apart in **anti-series**

$$v_A - v_B = e_k - e_{k+3} = 2 e_k$$

ideally double the voltage.

- Four MOSFETs with **cross-coupled gates**, so no gate driver is needed.
- They bypass the diodes, trading the fixed drop for a small R_{on} .
- Below $V_{off} \approx 1.2$ V it falls back to the diodes.

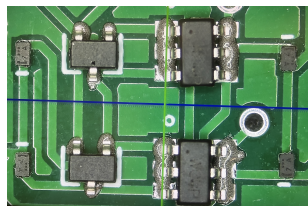
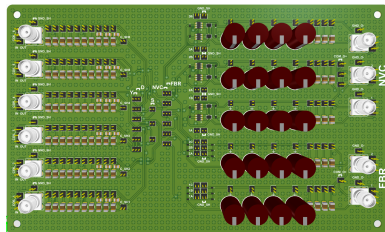


NVC for one anti-series pair.

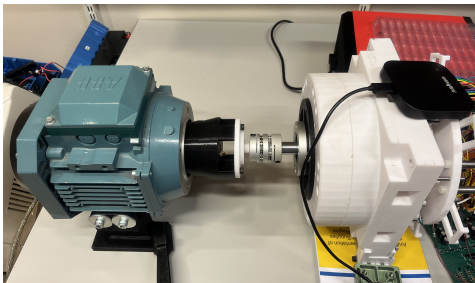
The prototype board



- Custom PCB, Altium Designer.
- Delta, Star and NVC selected by moving jumpers, plus a jumper bank for C_m .
- Hand assembled with soldering iron and hot plate.



One NVC bridge during assembly.



Motor coupled to the MP-VREH.



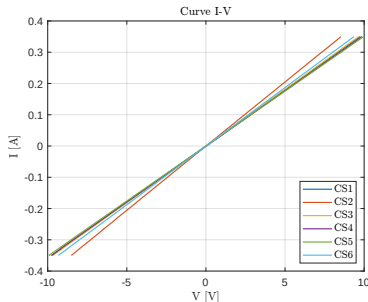
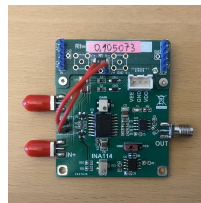
Acquisition device.



Programmable load.

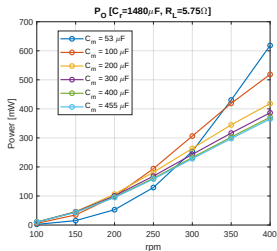
An inverter drives the motor and sets the speed precisely over the whole 100-400 rpm range.

- Six INA114AU boards, differential across a shunt.
- Shunts measured four-wire.
- Every sensor calibrated individually, into its own lookup table.
- Power trustworthy down to the milliwatt.

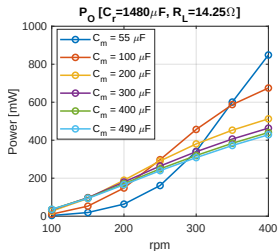


Calibration curves of the six sensors.

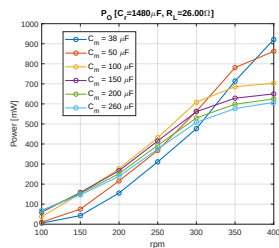
Simulation results



Delta

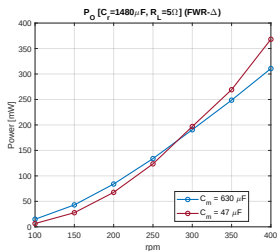


Star

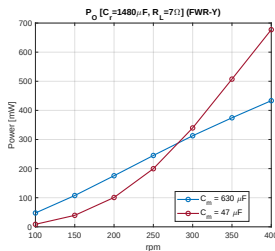


NVC

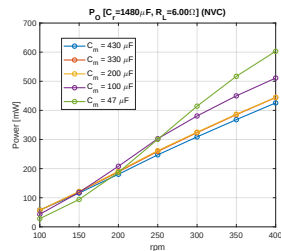
- Power against speed, for several fixed C_m .
- At 100 rpm, the worst case, the NVC gives **69.0 mW**, about $6\times$ the Delta (10.7) and nearly double the Star (36.7).
- The NVC is also the most efficient at every speed.



Delta



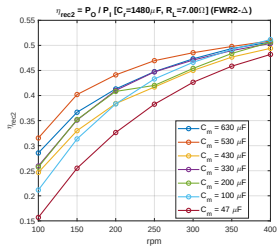
Star



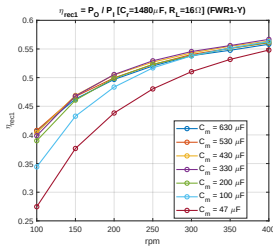
NVC

- At 100 rpm the NVC still wins with 57.7 mW, against 14.9 (Delta) and 47.6 (Star).
- Above 250 rpm the **Star overtakes the NVC**, 433.3 against 425.7 mW at 400 rpm.
- Simulation did not predict this crossover.

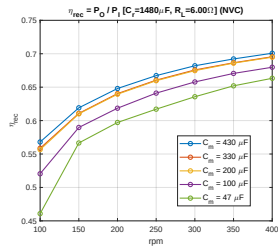
Measured efficiency



Delta



Star



NVC

- Best efficiency at every speed for the NVC, up to $\eta_{Rec} \approx 0.70$.
- Delta stops at ≈ 0.48 -0.51, the weakest as expected, and Star reaches ≈ 0.55 -0.57.
- The two rankings disagree, since efficiency says NVC while power says Star at high speed.

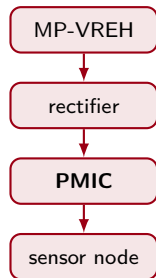
Measured power differs from simulation by 8% for the Star, up to almost 30% for the NVC.

	Delta	Star	NVC
Mean deviation of P_O from simulation	16%	8%	30%

Why?

- The ceramic C_m lose capacitance under the real voltage, which grows with speed.
- Component tolerances and the parasitic resistance of the bridges.
- Sources that are not perfectly sinusoidal.

PMIC	V_{in} (V)	Star up to
DS-AEM13920_VR	0.120-4.6	250 rpm
DS-AEM10941_VR	0.05-5	250 rpm
bq25570	0.1-5.1	250 rpm
SPV1050 (Boost)	0.15-5.3	250 rpm
SPV1050 (Buck-Boost)	0.15-18	full range
SPV1040	0.3-5.5	300 rpm



The PMIC turns the rectified power into a stable supply.

- Delta and NVC stay below 4 V and are **compatible with every chip tested**.
- The Star reaches 7 V and needs a wide-input converter or a clamp.

Conclusions

- **NVC and Star** are the two strongest candidates, one per end of the speed range; Delta is always the weakest.
- NVC and Delta fit any PMIC tested; the Star needs protection at high speed, its open-circuit voltage can damage the input.

Limitation

The harvester prototype is 3D-printed and sensitive to vibration at high speed, which makes the measurements harder to repeat.

Future work

Adaptive compensation tracking C_m with speed, and a full hardware test up to a real PMIC, to measure the efficiency from harvester to load.

Thank you for your attention!